The Anatomy of Scent

Process Book

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COM-480 Data Visualization, EPFL, Spring 2026

Overview

We built a scrollytelling website that explores what makes a perfume attractive. Using 24,063 perfumes scraped from Fragrantica and over 2,000 eBay listings, the site walks the reader through perfume composition patterns, gender differences in fragrance profiles, temporal trends spanning five decades, and the relationship between scent and market price. Eight interactive D3.js visualizations drive the narrative, each revealing a different facet of the data.

Our target audience sits at the intersection of three groups. Perfume enthusiasts who want to see their passion through a data lens come first. Then there are data visualization lovers who appreciate well crafted interactive storytelling. Finally, we wanted to reach curious people who have never considered why bergamot shows up in nearly every fragrance they own. The goal was to surprise all three.

The motivation behind this project came from a simple observation: perfumery is an art form built on hidden structure. Every perfume follows a three layer architecture of top, middle, and base notes. Certain ingredients appear thousands of times while others remain rare signatures. We wanted to make that invisible structure visible, and to ask whether the patterns we found could explain why some fragrances succeed while others fade into obscurity.

The site was designed to feel like walking into a high end perfume boutique. Dark background, refined typography, gold accents. Not a dashboard. A story, told section by section as the reader scrolls.

Data and Processing

Datasets

Our primary dataset is the **Fragrantica Fragrance Dataset** from Kaggle, containing 24,063 perfumes across 18 columns. Each entry includes the perfume name, brand, gender category, release year, user rating, fragrance accords, and three columns listing top, middle, and base notes as comma separated text. The gender distribution splits roughly 47% women, 32% unisex, and 21% men.

We supplemented this with the **Perfume E-commerce Dataset 2024**, also from Kaggle, containing around 2,000 eBay listings split between men's and women's perfumes. This dataset adds pricing information, with men's listings ranging from $3 to $259 (mean $46) and women's from $2 to $300 (mean $40). Since the data comes from eBay, the pricing insights reflect primarily the U.S. market.

Data Quality

The Fragrantica data was generally clean but required meaningful preprocessing. Fragrance notes were stored as plain text strings that needed parsing into structured arrays. Similar note names required grouping into broader families (floral, woody, citrus, spicy, and others). Release year was absent for 8.5% of entries, and user ratings were only available for a subset of perfumes with enough votes. The eBay data posed a different challenge: brand names did not always match cleanly between the two datasets, requiring fuzzy matching.

Processing Pipeline

We processed everything in Python using pandas. The pipeline parsed note text into arrays, grouped 500+ unique note names into families, computed co-occurrence matrices for the chord diagram, built the Sankey flow from top through middle to base notes, and aggregated statistics by gender and decade. A key decision was to pre-compute the expensive aggregations in Python rather than doing them client side. The raw Fragrantica CSV alone is over 12 MB, far too heavy for a browser to process interactively.

The pipeline outputs eight JSON files, each tailored to a specific visualization:

- `geo_data.json` (geographic note profiles by region)
- `notes_stats.json` (frequency and rating per note)
- `notes_stats_enhanced.json` (with family groupings)
- `temporal_trends.json` (note families by decade)
- `price_data.json` (eBay pricing merged with composition)
- `accords_data.json` (accord frequency and ratings)
- `chord_data.json` (pre-computed co-occurrence matrix)
- `sankey_data.json` (top to middle to base note flows)

This separation of concerns meant the frontend only needed to fetch small, purpose-built JSON files and bind them directly to D3.js selections. No client-side data wrangling.

Design Evolution

Initial Sketches

In Milestone 2, we sketched five core visualizations on paper. The opening section featured a beeswarm chart where each bubble represented a fragrance note, sized by frequency, with bubbles floating upward like scent molecules and clustered by note family. The second section placed side by side radar charts for men's and women's fragrance profiles. The third mapped note frequency against average rating in a bubble chart. A stacked area chart tracked note popularity across decades for the fourth section. The fifth combined eBay pricing with composition profiles in a scatter plot.

Two additional visualizations were planned as interactive sidebars: a chord diagram for note co-occurrence and a Sankey diagram for the top to middle to base flow.

Inspiration

The beeswarm concept drew directly from "Fragrance of Data," a project recognized at the Information is Beautiful Awards that used a similar particle aesthetic to represent scent data. Our scrollytelling structure was inspired by The Pudding, whose long form data stories use scroll position to control narrative pacing. We wanted the same feeling of the data revealing itself gradually as the reader moves through the page.

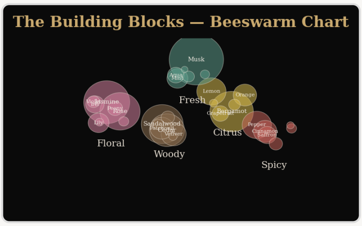
Visual Identity Decisions

Early on, we committed to a dark luxury aesthetic that would set the tone before any data appeared. The color scheme uses `#0a0a0a` as the primary background with `#c9a96e` gold accents, a palette that evokes the matte black and gold foil of high end perfume packaging. Typography pairs Cormorant Garamond for headings (an elegant serif with visible calligraphic DNA) with DM Sans for body text (a clean geometric sans serif that reads well at small sizes on dark backgrounds).

How the Sketches Evolved

BEESWARM (SECTION 1)

M2 sketch: Static bubble chart with manual positioning

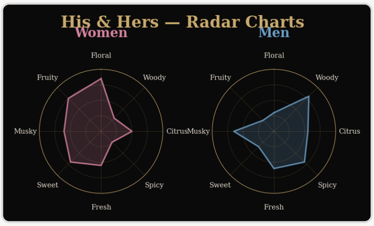


. All notes visible at once, creating visual clutter with 500+ circles competing for attention.

Final: D3 force simulation with scroll triggered transitions. The chart reveals notes progressively across four scroll steps: first musk alone, then bergamot, then the top 10, then the full constellation. Family clustering uses color encoding rather than spatial separation.

RADAR CHARTS (SECTION 2)

M2 sketch: Two separate static radar charts placed side by side

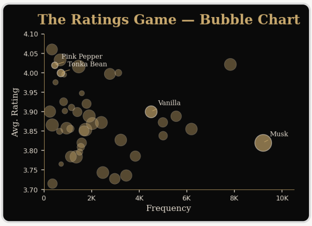


. Eight axes representing the top note families.

Final: Overlaid radar polygons that animate in sequence. Women's profile appears first, then men's overlays with a different color, making the comparison immediate. A toggle adds unisex as a third layer. The axes label the actual note families from the data rather than generic categories.

BUBBLE CHART (SECTION 3)

M2 sketch: Simple scatter plot with uniform bubble sizes

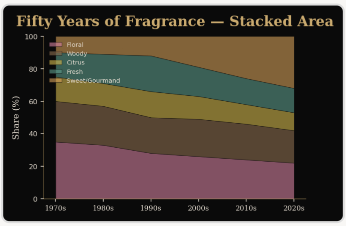


. Frequency on x-axis, rating on y-axis.

Final: Log-scale x-axis for better spread of common vs. rare notes. Bubble size encodes total perfume count. Step-driven highlighting dims non-relevant notes and adds a glow filter to emphasized ones. Labels for the top 10 notes prevent the reader from needing to hover for context.

TIMELINE (SECTION 4)

M2 sketch: Simple stacked area chart from 1970 to 2020

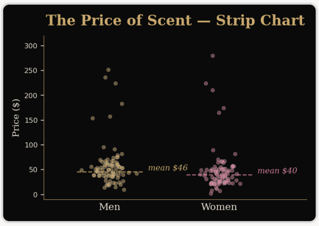


, showing all note families simultaneously.

Final: Scroll controlled decade highlighting. As the reader progresses, each decade receives a focus state with annotations explaining the cultural context (the clean 1990s, the gourmand 2010s). The non-focused decades fade to lower opacity.

PRICE STRIP (SECTION 5)

M2 sketch: Basic strip chart of price points by gender



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Final: Full beeswarm dodge algorithm to prevent dot overlap. Dots sized by units sold, colored by fragrance type (EDP, EDT, Parfum, EDC). Mean price lines with annotations. Three scroll steps: overview, premium highlight ($100+), and brand cluster labels for 10 major brands.

Key Technical Decisions

Scrollama wraps the IntersectionObserver API to determine which narrative step is active, letting us define scroll beats as HTML divs and trigger D3 transitions declaratively. We initially used **Lenis** for smooth scrolling but removed it after it caused lag with D3 force simulations, so native CSS `scroll-behavior: smooth` proved sufficient. We centralized tooltip logic, color scales, and note family mappings in `main.js`; each of the eight visualization modules follows the same pattern: one init function, one JSON file, one SVG. Co-occurrence and Sankey computations were moved to Python after the JavaScript version took 4+ seconds.

> CHORD + SANKEY (SECTION 6) & HEATMAP (SECTION 7)
>
> **Chord & Sankey:** Originally planned as floating sidebars, moved to a dedicated two-column card section after z-index conflicts with sticky containers.
>
> **Heatmap:** Replaced the planned perfume search feature. The heatmap proved more visually striking and told a clearer story about gender differences, while requiring far less data (no individual perfume records).

Implementation and Challenges

Technology Stack

Layer	Technology	Purpose
Visualization	D3.js v7	All eight interactive charts
Scroll management	Scrollama	Triggering transitions on scroll position
Sankey layout	d3-sankey 0.12	Node/link positioning for flow diagram
Data processing	Python 3, pandas	CSV parsing, aggregation, JSON export
Frontend	Vanilla HTML/CSS/JS	No framework, no build step
Smooth scroll	CSS native	Replaced Lenis (caused lag with D3 simulations)
Hosting	Vercel	Static deployment from `/docs`

We deliberately avoided frontend frameworks. A scrollytelling site with eight D3 visualizations does not need React's component model, and the absence of a build step means faster iteration and simpler deployment. Every visualization lives in its own file under `js/visualizations/`, and `main.js` orchestrates initialization, scroll events, and shared utilities like color scales and tooltip rendering.

Architecture

The site follows a simple pattern. The HTML defines eight section containers, each with an `id` matching its visualization (e.g., `viz-beeswarm`, `viz-chord`). The five scrollytelling sections use a two column layout: a sticky graphic container on the left holds the SVG, while scrollable text steps on the right drive transitions. The remaining three visualizations (chord, sankey, heatmap) use simpler layouts since they do not need scroll linked narrative.

Data flows in one direction: Python scripts read the raw CSVs, compute aggregations, and write JSON files to `docs/data/`. The JavaScript never touches raw data. Each visualization module fetches its JSON, binds the data to SVG elements using D3's data join pattern, and registers its own interaction handlers.

Key Challenges

Note taxonomy. Over 500 unique note names with no standard taxonomy. We built a manually curated mapping into seven families (floral, woody, citrus, spicy, fresh, sweet, musky) based on perfumery references, giving a consistent lens across all visualizations.

SVG sizing bug. Sticky containers inside flex parents had zero computed width on page load. The fix was lazy initialization via IntersectionObserver, where each visualization only initializes when its container enters the viewport.

Dark theme contrast. Grid lines at opacity 0.06 were invisible on most monitors. Three rounds of tuning: grid lines to 0.15, axis text to `#9a9088`, subtle gold borders on card containers.

Responsive design. CSS media queries stack the layout vertically below 768px, replacing the sticky side-by-side pattern. The experience is optimized for desktop.

Deployment. The site deploys from `/docs` via Vercel with no build step. Total deployed size is under 2 MB of pre-processed JSON.

Key Insights

> ***Musk and bergamot are universal.*** *These two notes appear in both men's and women's top 5 most frequent ingredients, forming the invisible foundation of modern perfumery across all gender categories.*

> ***Rarity correlates with quality.*** *Notes like pink pepper and tonka bean appear in fewer perfumes but carry the highest average ratings. Meanwhile, gender differences are real but concentrated: floral notes dominate women's compositions, woody and spicy notes dominate men's, but the overlap is larger than expected.*

> ***Composition trends mirror culture; price does not mirror composition.*** *The timeline reveals a clear arc from aldehydic glamour (1970s) through clean aquatic (1990s) to gourmand and oud (2010s–2020s). Yet the price scatter shows no clean correlation between ingredient sophistication and market value. Brand cachet drives pricing more than raw composition.*

> ***Geography shapes scent identity.*** *An interactive world map with radar-on-hover (added after M2 feedback) reveals that the Middle East stands apart from every other region. Higher woody, spicy, and musky proportions reflect an oud-based tradition fundamentally different from the Western floral canon. Europe and North America overlap heavily, while Asia favors lighter, citrus-forward profiles.*

Peer Assessment

We split the work across three team members, each owning distinct deliverables.

Area	Alexandre Mourot	Gaël Conde Losada	Rayane Charif Chefchaouni
Data	Processing pipeline, JSON generation, note taxonomy	Exploratory data analysis, dataset selection	Geographic data preprocessing, region mapping
Architecture	Website structure, Scrollama setup, lazy init	Milestone reports (M1, M2), process book	World map integration, TopoJSON pipeline
Visualizations	Beeswarm, radar, bubble chart, heatmap	Chord, Sankey, timeline, price strip	Geographic world map, geographic radar
Design	CSS theme, responsive layout, typography	Narrative structure, scroll step content	UX interaction hints, screencast

What We Would Do Differently

Earlier user testing would have caught issues like invisible grid lines and confusing axis labels sooner. **A perfume search feature** (dropped in favor of the heatmap) would have made the site more personal. **A dedicated mobile layout** with simplified charts would serve phone users better. And a **larger pricing dataset** (the eBay data covers only ~2,000 listings) would have strengthened our weakest section.

Conclusion

The Anatomy of Scent transforms 24,063 rows of perfume data into a story anyone can follow. Through eight interactive visualizations, including a world map added in response to M2 feedback, we made the hidden structure of perfumery visible and explorable across composition, gender, time, geography, and price. The project taught us that the hardest part of data visualization is not the code or the data. It is choosing what to show, when to show it, and how much to let the reader discover on their own.